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Abstract

Conversational generative artificial intelligence (AI) is positioned to benefit youth mental health significantly. Chatbots, virtual assistants, and other conversational generative AI can play a key role in easing the burden on services and clinicians, positively impacting mental health outcomes for clients, and supporting real-time, proactive, shared, and appropriate treatment decisions. This research aims to determine the design and implementation requirements of a novel conversational generative AI for youth mental health, Mental health Intelligence Agent (Mia). This work reports preliminary findings from co-design workshops with internal stakeholders. Future co-design work with young people, health professionals, and other stakeholders will further inform system development, driven by stakeholder feedback to ensure safety, acceptability, usability, and functionality. Furthermore, the breadth of multi-stakeholder, multi-site research data to be collected will draw out findings with wider implications for future research and development, policy-making, and real-world digital technology design practice in the youth mental health space.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; • **Information systems** → Information systems applications; Decision support systems; • **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods.

Keywords

Artificial intelligence, Chatbot, Co-design workshop, Prototyping, User experience

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1 Introduction

With the advent of ChatGPT in 2022, the generative artificial intelligence (AI) field has received significant attention, propelling it into the mainstream [1]. Generative AI is now applied widely, spanning a broad range of application domains, such as creative arts, customer service, architectural visualisation, fashion design, drug discovery and molecular design, and video game design [16].

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These systems are implemented to support various functions, including image generation, programming code generation, copywriting, translation, and conversational human-computer interaction [17, 33]. Perhaps the most commonplace, public-facing generative AI are those that are conversational agents, typically implemented as voice assistant smart devices, virtual assistants, or chatbots [18]. Conversational generative AI systems combine conversational AI and generative AI functionality, using natural language processing and machine learning techniques, such as large language models (LLMs), to participate in human-like conversations with users, classify large volumes of text, answer questions, and translate languages and incorporate generative AI capabilities to generate new text based on some input beyond standardised, untailored responses [18].

Research indicates that conversational generative AI in mental health care is positioned to alleviate the burden on services and deliver various benefits for mental health professionals and clients [4, 15]. Compared to conversational AI systems alone, generative AI-enabled systems demonstrate more positive effects on mental health outcomes [22]. Owing to substantial data collection and processing capabilities and context-driven adaptability, conversational generative AI is projected to play a key role in supporting personalised, scalable, and accessible mental health care [2, 21, 29]. A 2019 review of the literature on conversational agents in psychiatry and their clinical roles indicates that they may benefit psychoeducation and self-adherence [32]. Furthermore, the review found that users typically report high satisfaction and low risk of harm with conversational agents used in this context [32]. In a 2024 study, researchers evaluated the ability of four prevalent LLMs to assess schizophrenia prognosis and found that the assessments of three out of the four LLMs evaluated aligned with those of mental health professionals used as a comparison [10]. A study in Malawi that evaluated the effectiveness of a mental health chatbot mobile application reported reduced scores on measures of depression, anxiety, and burnout, improved resilience scores, and increased frequency of resilience-building activities among participants using the intervention [20]. Positioned in a complementary role in clinician-client collaboration, conversational generative AI has been shown to effectively assist in empathic clinician documentation and patient-facing interactions, potentially indicating a key function in combating compassion fatigue among clinicians [4]. Research highlights that conversational generative AI can be utilised to support clinical decision-making by, for instance, generating accurate lists of differential diagnoses to overcome complex diagnostic challenges [4] and supporting non-experts to communicate and understand complex health concerns and information and subsequently make informed decisions [13].

Globally, mental illness presents a critical health issue for young people aged 12–25 years [23]. Consequently, youth mental health services are overburdened [23]. Furthermore, there exists a wide spectrum of mental illnesses in presenting populations, adding to the difficulty of addressing all clients' needs and risking leaving some clients without care [25]. The state of youth mental health often leads to delayed, reactive, and inappropriate treatment decisions [24, 25]. Digital technologies present an opportunity to alleviate these concerns [8] and further support shared and informed decision-making by enhancing the breadth of information

considered to best treat the myriad mental illnesses in presenting populations in youth mental health services [3, 6].

Yet, the design, development, implementation, evaluation, and governance of conversational generative AI require meaningful consideration of underlying key determinants that shape user experiences to ensure intervention effectiveness [22] (e.g., therapeutic relationships that incorporate AI systems, effective communication, content quality and engagement); user preferences and requirements relating to usability and trust [28, 30, 35]; relevant regulation to protect users from risk of harm [9]; and the diversity of unmet needs among users, particularly those from vulnerable communities, in mental health that might be exacerbated by otherwise non-inclusive practice [5, 11, 12, 27]. This necessitates the interdisciplinary co-design of conversational generative AI in mental health.

This research aims to determine the design and implementation requirements of a novel conversational generative AI for youth mental health, Mental health Intelligence Agent (Mia). This work reports preliminary findings from co-design workshops with internal stakeholders representing mental health and digital technology expertise. The findings advise the development of a minimum viable product (or prototype) and proposed future co-design work with young people, mental health professionals, and other stakeholders.

2 Initial Development

2.1 Study context

The work reported here is part of a larger project exploring the use of AI to scale the availability of mental health services for the targeted populations while still delivering personalised treatment decisions in youth mental health services. Principally, the project seeks to produce a suite of intelligence and analytics digital technologies to support shared and informed decision-making in youth mental health services, aiming to address the often delayed, reactive, and inappropriate treatment decisions in youth mental health emerging from issues relating to overburdened services and the inherent heterogeneity of mental illness in presenting populations. The digital technologies examined in the project prioritise advancing personalised, dynamic (measurement-based), and proactive youth mental health care. One of those digital technologies being explored is a conversational generative AI for youth mental health, Mia.

2.2 Mental health Intelligence Agent (Mia)

At the time of writing, Mia is in development. Principally, Mia is an AI expert agent that uses an LLM and reasoning engine to scale expertise with specific knowledge about youth mental health related to the Brain and Mind Centre Youth Model of Care [14, 26]. The final design product intends to be capable of interpreting user inputs, accessing relevant knowledge banks, and providing tailored, explainable responses about a young person's mental health, specifically for early-stage mood disorders. Chiefly, the intended users are practising mental health professionals operating in service settings, although clients (young people and their families) are also being explored as potential users. Figure 1 illustrates the basic user interaction and system context for Mia.

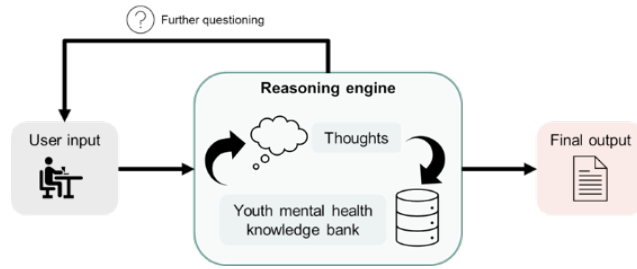


Figure 1: Combined basic user interaction and system context diagram for Mia.

To follow is a list of the key capabilities to be incorporated into Mia:

- Translates a person’s mental health history into personalised summaries, a clinical formulation, and a treatment plan to help clinicians identify the next steps for assessment and care planning.
- Accepts user input to query what is known about a person and dynamically updates recommendations based on the newest information.
- Accesses a database of curated evidence-based literature, best practice clinical guidelines, and expert knowledge (also referred to as the ‘youth mental health knowledge bank’).

A local technology partner, UNCAPT, is developing Mia. The database used to train and support Mia incorporates state-of-the-art research collated by the research team (i.e., published journal articles on youth mental health concepts), various outputs from the research team (i.e., mental health podcast transcripts from an expert clinician), and prompting questions for Mia to pose to users to stimulate more user input (i.e., requests for more information relating to critical signs to assess for in young people, principled multidimensional assessment, and confidence threshold criteria for developing a treatment plan). Comprehensively describing the extensive development and evaluation processes for Mia’s underlying systems is out of scope here; this work focuses exclusively on the co-design of Mia’s user interface (UI), possible use, and expected implementation.

3 Methodology

Three two-hour preliminary workshops were held with internal stakeholders throughout 2023 and 2024. Stakeholders were those on or associated with the Youth Mental Health and Technology team at the University of Sydney’s Brain and Mind Centre and the technology partner’s team. The workshops aimed to determine the design and implementation requirements of Mia and ultimately advise the development of a prototype that will later be explored, advanced, and evaluated with external health professionals, thus reducing the burden on this participant group in future research. Crucially, this process enables internal knowledge and expertise to be utilised but does not situate the internal stakeholders as the subject of the research. Data was collected in the form of workshop recordings, stakeholder artefacts, and field notes and used to contribute to iterative research and development processes only.

This research was approved by the University of Sydney Human Research Ethics Committee (Protocol no: 2021/680).

3.1 Fine-tuning workshop

In November 2023, a 3-hour ‘fine-tuning’ workshop was held. The purpose of the workshop was to (1) enable expert stakeholders to interact with an early Mia prototype, (2) comment on Mia outputs and make suggestions to better align Mia’s reasoning with their expertise, and (3) discuss user experience, usability, utility, potential uses and users (including clients), recommended system changes, and perceived acceptance within mental healthcare. At this stage in development, Mia was a simple chatbot capable of handling question-answer interactions.

Two use cases were presented to the stakeholders: (1) Mia used by mental health professionals and (2) Mia used by young people and their families. In the first use case, Mia was positioned to support clinicians in developing a clinical formulation and hypothesis generation (i.e., possible diagnosis, illness trajectory, and clinical stage), identifying the next steps for assessment or intervention, and identifying multiple personalised treatment options. In the second use case, Mia was placed in the role of supporting a young person or family member to learn more about their own (or a family member’s) mental health, identify critical questions to take to a health professional (empowerment), become informed about a range of treatment options available to them, and learn strategies for self-management.

3.2 Co-design workshops

In April 2024, two co-design workshops were held with seven stakeholders (four in the first and three in the second). Stakeholders included psychiatrists, general and clinical psychologists, youth mental health services managers, and a technology partner representative. The workshops aimed to engage in a richer set of co-design activities and discussions to refine and finalise Mia’s system requirements for prototype development. At this stage, Mia was conceptualised as both a chatbot and an electronic medical record system. The research team met on numerous occasions to reflect on the new vision for Mia and explore new features and UI mockup options. Figure 2 features one UI mockup explored during one of those research team meetings.

To date, the projected capabilities of Mia have progressed to recording multiple clients’ information, translating unstructured user input about individual clients’ mental health history into personalised summaries, recommending future assessment and care planning, and handling question-answer interactions about specific clients in addition to generalised mental health expert information.

Unlike the first fine-tuning workshop, the two co-design workshops had explicit themes, which intended to focus the attention of the stakeholders to inform the prototype development to follow. The themes were directly communicated to the stakeholders during the workshops. The themes included: (1) user experience and user flows, (2) usefulness in helping clinicians achieve goals, and (3) issues, improvements, barriers, and challenges. The planned workshop structure was as follows:

- Introduction to project and co-design workshop purpose.

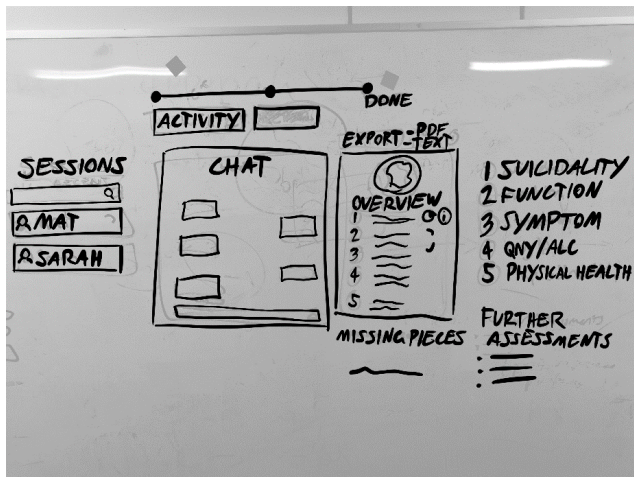


Figure 2: Example Mia UI mockup drawn during a team meeting prior to the workshops. Digital illustration overlaid on photograph.

- Presentation and walkthrough of Mia mockups, use cases, user flows, intended users, and environment.
- First co-design activity.
- Group discussion.
- Second co-design activity.

Planned co-design activities included paper prototyping to generate stakeholder-led UIs and user flows and design cards with questions to prompt stakeholder-led discussion. Group discussion was prompted by questions relating to Mia's proposed features, stakeholders' requested features, system interoperability, design considerations to improve user experience, utility, and usability, and emerging issues, barriers, challenges, or improvements. Unlike the fine-tuning workshop, the stakeholders did not engage with Mia directly.

Physical materials were supplied to the stakeholders to provide written feedback, including A2 paper (or butchers paper) and printed copies of the presentation. Written feedback was encouraged to capture stakeholders' key shared points and to provide stakeholders with an alternative method to contribute outside the group discussion. The butchers paper was also intended to facilitate paper prototyping. A follow-up survey was distributed to stakeholders to provide additional opportunities to get feedback on different UI design options.

Two use cases were discussed in the workshops, both concerning mental health professionals as the user to: (1) generate a personalised summary of a client's mental health needs based on an unstructured interview and notes gathered during an intake assessment and (2) get an update on a returning client before an upcoming appointment and add new information based on the appointment.

4 Results and Discussion

Basic inductive qualitative analysis and system requirements analysis [7] were conducted to determine design and implementation

requirements of Mia. Feedback from the fine-tuning workshop comprised of group discussions and field notes. Stakeholders reflected on their interactions with the early Mia prototype and discussed various topics relating to user experience, adoption, implementation, and system functionality requirements. Initial feedback was that the threshold for generating a final output was quite high; many stakeholders recounted having to respond to many follow-up questions requesting additional user input before Mia would produce useful recommendations for possible next steps for mental health assessment or intervention and personalised treatment options. Stakeholders noted that although a high threshold was beneficial for developing a comprehensive mental health assessment, it would not be appropriate for practical use in environments where clinicians are time-poor. To balance practicality and efficacy, a consensus was reached on requiring a lower threshold for Mia to produce a final output, limiting requests for additional user input to a small number of critical follow-up questions. As reported in the literature, creating this balance is essential to avoiding additional burdens and time-consuming tasks for mental health clinicians and other system users [31]. In a group discussion about young people (as mental healthcare clients) using Mia, the stakeholders agreed that young people directly interacting with Mia in a longer back-and-forth conversation would be an interesting use case to explore going forward, as it would allow them to input their information and tell their story in their own words and have it summarised and contextualised by the knowledge bank. However, more research is needed to understand related risks, such as potential overreliance [19].

Most stakeholder feedback from the co-design workshops consisted of group discussions, field notes, and written feedback on printed copies of the presentation (see Figure 3 for examples). The planned co-design activities—paper prototyping for UI design and user flow feedback and design cards to prompt discussion—were not conducted for several reasons. First, the stakeholders' written feedback on the printed copies of the presentation, which included several UI mockup options for Mia (see Figure 3), generated adequate feedback for designing the UI. Second, the group discussion created sufficient feedback for informing the user flows. Last, the design cards were not used as the group discussion questions posed by the facilitators effectively prompted discussion. The butchers paper was underutilised as the planned co-design activities were not completed, given the richness of group discussion. However, providing butchers paper supported stakeholders in providing summaries of their key shared ideas at the end of the workshops (see Figure 4).

Following system requirements analysis, key use cases, UI views, and functional requirements were identified. Stakeholders regarded three use cases appropriate for further consideration in the next Mia prototype iteration: (1) a clinician adds a new client, (2) a clinician adds more information about an ongoing client, and (3) a young person adds information about themselves indirectly to Mia (i.e., they do not have access to any Mia responses or outputs). These use cases were further developed as user flows with step-by-step instructions describing a typical user interaction (e.g., user inputs, system navigation, system outputs) and what UI views are involved in the user flows. Mia requires multiple UI views, as informed by stakeholder feedback. Table 1 identifies those views and details

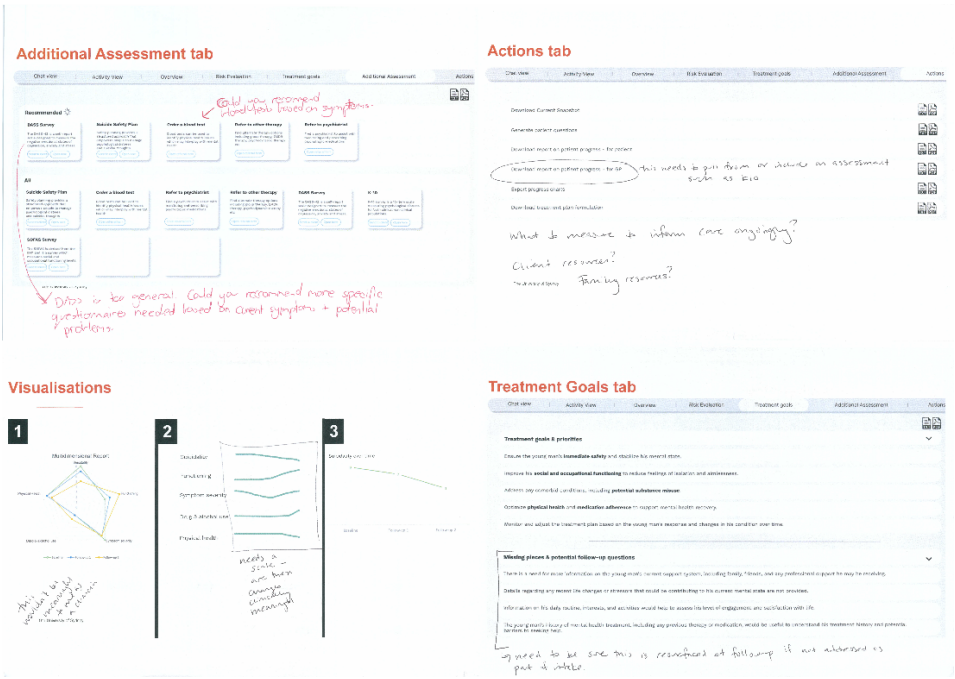


Figure 3: Examples of written feedback on printed copies of the presentation.

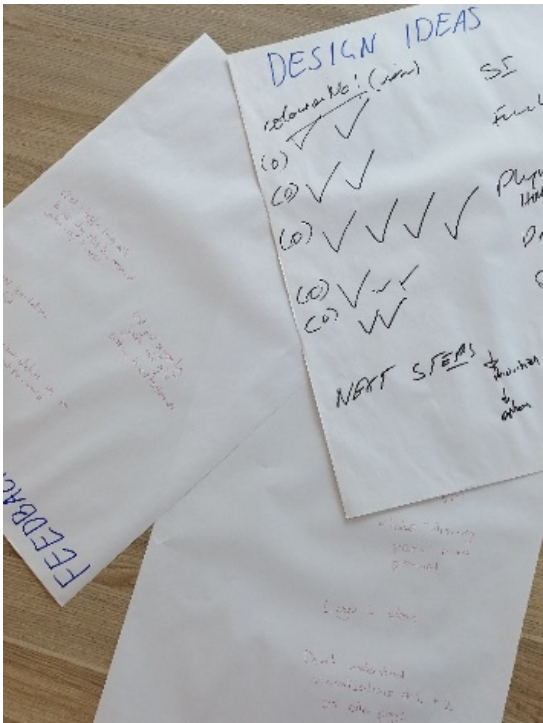


Figure 4: Examples of stakeholder feedback on butchers paper from the co-design workshops.

the related functionality of each view. Stakeholders determined numerous key functional requirements for Mia, detailed in Table 2.

At the conclusion of the preliminary research reported here, the findings were presented in a system requirements document [34] to guide the Mia prototype development. The document was delivered to and revised in collaboration with the technology partners to ensure a shared understanding of the vision for Mia and to revise priorities.

5 Conclusions and Future Work

The next steps for this project are conducting further co-design workshops and interviews with young people (and their friends and families) and practising mental health professionals. Future workshops aim to enable stakeholders to interact with Mia and create a rich understanding of a variety of factors relating to Mia. These include user experience, clinical utility, impact, feasibility, adoption, service implementation, user groups, user requirements, and ethical considerations. Several key risks will be explored with stakeholders to collaboratively design user-centred solutions that minimise or prevent them, including the potential added burden on clinicians and clients from time-consuming chatbot interactions, frequent data input, overreliance on the system, and any subsequent negative impacts on users.

Expected future results will further inform the development of Mia, driven by direct feedback from these key stakeholders to ensure safety, acceptability, usability, and functionality. Furthermore, the breadth of multi-stakeholder, multi-site research data to be collected will draw out findings with wider implications for future research and development, policymaking, and real-world digital technology design practice in the youth mental health space.

Table 1: Required UI views and related functionality for Mia

User Interface View	Functionality
Login	Requests login details as either a clinician or client user.
Intake Chat	Allow users to input new client information. Mia prompts for more input to create an initial report sufficiently.
Ongoing Chat	Allow users to add more information about an ongoing client (clinician user) or add more information about themselves (client user). Mia outputs should be non-diagnostic.
Activity	Displays underlying Mia processes and logic.
Overview	Displays summaries and visualisations relating to client information.
Next Steps	Displays possible treatment goals and additional clinical assessments relating to a client generated by Mia.
Client Timeline	Displays a timeline of changes in client state.
Actions	Displays outputs that the user can generate as files.

Table 2: Key functional requirements for Mia

Functional Requirement	Description
System confidence indicator	Mia should determine system confidence, indicating that enough information has been gathered to surpass a confidence threshold and generate recommendations reflective of an expert clinician's recommendations.
Establish <i>knowns</i> and <i>unknowns</i>	Mia should generate key information it knows and does not know about a client.
Inferred symptom validation check	Mia should prompt the user to validate inferences made about client symptoms from user input.
Multiple data inputs	Mia should accept document and audio file inputs to add to client information.
Generate <i>next steps</i>	Mia should generate possible treatment goals, additional clinical assessments, and interventions.
Detect and monitor risk to a client	When a potential risk to a client is identified, Mia should inform the user of the possible risk and provide immediate options to address the risk. Mia should track changes in risk over time.
Client timeline	Mia should generate and visualise a client's history of clinical assessment scores and key events.
Outputs	Users should be able to generate various reports/resources in document formats.

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